

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi. \quad (3.1)$$

The Noether current is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \rightarrow \psi' = e^{i\alpha}\psi \quad (3.2)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = e^{-i\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \quad (3.3)$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \rightarrow \alpha(x). \quad (3.4)$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ :

$$D_\mu\psi(x) = (\partial_\mu - igA_\mu)\psi(x). \quad (3.5)$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \rightarrow A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \quad (3.6)$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \rightarrow (D_\mu\psi)' = \partial_\mu\psi' - igA'_\mu\psi' = e^{i\alpha}D_\mu\psi. \quad (3.7)$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \quad (3.8)$$

$$= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \quad (3.9)$$

For photon dynamical, we need to add kinetic term for A_μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (3.10)$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 $SU(N)$ gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j , where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^N i\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \quad (3.11)$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \rightarrow \psi'_j = \sum_k U_{jk} \psi_k, \quad (3.12)$$

$$\bar{\psi}_j \rightarrow \bar{\psi}'_j = \sum_k \bar{\psi}_k U_{kj}^\dagger, \quad (3.13)$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \rightarrow \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U_{kj}^\dagger U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \quad (3.14)$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \rightarrow U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - ig \sum_k (A_\mu)_{jk} \psi_k. \quad (3.15)$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \rightarrow (D_\mu \psi_j)' = \partial_\mu \psi'_j - ig \sum_k (A'_\mu)_{jk} \psi'_k \quad (3.16)$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \quad (3.17)$$

For convenient, we will use the short-hand notation(Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = U D_\mu \psi = e^{i\alpha(x)} D_\mu \psi. \quad (3.18)$$

We claim that the gauge field transforms following

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.19)$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = \partial_\mu (U\psi) - ig \left(U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger \right) U\psi \quad (3.20)$$

$$= U \partial_\mu \psi + (\partial_\mu U) \psi - ig U A_\mu \psi - (\partial_\mu U) \psi \quad (3.21)$$

$$= U (\partial_\mu \psi - ig A_\mu \psi) = U D_\mu \psi. \quad (3.22)$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as “non-Abelian”, but until now it’s not quite obvious to see how it is “non-Abelian”, therefore, the field strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \quad (3.23)$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - ig[A'_\mu, A'_\nu] = U F_{\mu\nu} U^\dagger. \quad (3.24)$$

The Lagrangian is proportion to the trace over $F_{\mu\nu} F^{\mu\nu} = (F_{\mu\nu})_{jk} (F^{\mu\nu})^{kj} = \text{tr}(F_{\mu\nu} F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\text{tr} F_{\mu\nu}$ transfers as

$$\begin{aligned} \text{tr}(F_{\mu\nu} F^{\mu\nu}) &\rightarrow \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) = \text{tr}(U F_{\mu\nu} U^\dagger U F^{\mu\nu} U^\dagger) \\ &= \text{tr}(U U^\dagger F_{\mu\nu} F^{\mu\nu}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}). \end{aligned} \quad (3.25)$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}) + \sum_{j,k=1}^N [\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m \bar{\psi}_j \psi_j], \quad (3.26)$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it’s necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{i\alpha} \simeq 1 + i\alpha. \quad (3.27)$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \quad (3.28)$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \quad (3.29)$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad (\text{anti-hermitian}) \quad (3.30)$$

where we note $\hat{H} = iH$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = i \sum_a \alpha^a (t^a)_{jk} \quad (3.31)$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are $\mathbf{N}^2 - \mathbf{1}$ **generators in $\mathbf{SU}(\mathbf{N})$** . Further more, base on $\det U = 1$ and the Jacobi identity $\det(e^M) = e^{\text{tr}(M)}$, we find

$$\det U = \det(e^{\hat{H}}) = e^{\text{tr} \hat{H}} = 1, \quad (3.32)$$

thus, $\text{tr} \hat{H} = 0$ and implies

$$\text{tr}(t^a) = 0. \quad (3.33)$$

So t^a are **traceless generators**.

Now we discuss the algebra structure for generators t^a . The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = i \sum_c f^{abc} t^c. \quad (3.34)$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^d \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \quad (3.35)$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b . Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \dots, \alpha_d; 0, \dots, 0) = \gamma_a(0, \dots, 0; \alpha_1, \dots, \alpha_d) = \alpha_a \quad (3.36)$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \dots, \alpha_d) \cdot g(\beta_1, \dots, \beta_d) = g(\gamma_1, \dots, \gamma_d), \quad (3.37)$$

in which γ is the function of α and β :

$$\gamma_a = \gamma_a(\alpha_1, \dots, \alpha_d; \beta_1, \dots, \beta_d). \quad (3.38)$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C_{bc}^a \alpha_b \beta_c + \dots, \quad (3.39)$$

where C_{bc}^a is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} [(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b] + \dots, \quad (3.40)$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + i \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \dots \quad (3.41)$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + i \sum_{a,b,c} C_{bc}^a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \dots \quad (3.42)$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = i \sum_c C_c^{ab} t^c. \quad (3.43)$$

this gives the structure constant:

$$[t^a, t^b] = i \sum_{c=1}^d (C_c^{ba} - C_c^{ab}) t^c \quad (3.44)$$

$$= i \sum_{c=1}^d f^{abc} t^c \quad (3.45)$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.

Examples of Lie groups

- **GL(N)**: Group of $N \times N$ matrices in $\mathbb{R}^N$ with $\det M \neq 0$.
- **U(N)**: Group of unitary $N \times N$ matrices in $\mathbb{C}^N$, satisfies

$$U^\dagger U = U U^\dagger = \mathbb{1} \quad \text{and} \quad \det U \neq 0. \quad (3.46)$$

Infinitesimal version:

$$U = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a. \quad (3.47)$$

with $d = N^2$ is the dimension of $U(N)$ Lie algebra/group.

- **SU(N)** := $\{A \in U(N) | \det A = 1\}$. Dimension of Lie algebra $\mathcal{SU}(N)$ is $d = N^2 - 1$. Generators t^a are traceless

$$\det U = \det \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) = 1 + i \sum_{a=1}^d \alpha_a \text{tr } t^a = 1 \quad (3.48)$$

thus

$$\text{tr } t^a = 0 \quad (3.49)$$

- **O(N)**: group of orthogonal $N \times N$ matrices in $\mathbb{R}^N$.

$$O O^T = O^T O = \mathbb{1} \quad (3.50)$$

Infinitesimal version:

$$O = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \quad (3.51)$$

Dimension of Lie algebra $\mathcal{O}(N)$ is $\frac{1}{2}N(N-1)$. And notice:

$$O O^T = \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) \left(\mathbb{1} + i \sum_{b=1}^d \alpha_b (t^b)^T \right) = \mathbb{1}, \quad (3.52)$$

thus,

$$\mathbb{1} = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + i \sum_{b=1}^d \alpha_b (t^b)^T + \mathcal{O}(\alpha^2) \quad (3.53)$$

$$= \mathbb{1} + i \sum_{a=1}^d (\alpha_a t^a + \alpha_a (t^a)^T), \quad (3.54)$$

which implies

$$t^a = -(t^a)^T. \quad (3.55)$$

- $\mathbf{SO}(\mathbf{N}) := \{A \in O(N) \mid \det A = 1\}$
- $\mathbf{SO}(\mathbf{N}, \mathbf{M}) = \{\Lambda \in \text{Mat}(N+M, \mathbb{R}) \mid \Lambda^T \eta \Lambda = \eta, \det \Lambda = 1\}$, where

$$\eta = \text{diag}(\underbrace{1, \dots, 1}_N, \underbrace{-1, \dots, -1}_M) \quad (3.56)$$

- **Lorentz group:** $SO(1, 3)$

1.4 Representations of Lie algebra

Def: Representation of group

A **representation of a group G on a d_r -dimensional vector space V** is a group homomorphism

$$\rho : G \rightarrow GL(V), \quad (3.57)$$

which maps each group element $g \in G$ to an invertible linear operator $\rho(g)$ acting on V , such that:

$$\rho(g_i) \cdot \rho(g_j) = \rho(g_i g_j) \quad \forall g_i, g_j \in G. \quad (3.58)$$

The dimension d_r of the vector space V is called **the dimension of the representation**.

- A representation ρ is called **reducible** if there exists a non-trivial proper subspace $U \subset V$ (where $U \neq \{0\}$ and $U \neq V$) that is invariant under the action of all group elements, meaning:

$$\rho(g)u \in U \quad \forall g \in G, \forall u \in U. \quad (3.59)$$

- If no such invariant subspace exists, the representation is called irreducible (irrep).

For a Lie group characterized by continuous parameters α , a representation is a mapping $\rho : g(\alpha) \rightarrow M(\alpha)$, where the matrices (or operators) $M(\alpha)$ preserve the group multiplication:

$$M(\alpha) \cdot M(\beta) = M(\gamma) \quad \text{whenever} \quad g(\alpha) \cdot g(\beta) = g(\gamma). \quad (3.60)$$

In the next, we introduce some useful representations.

- **Fundamental(defining) representation**

For a matrix Lie group, the **fundamental representation** is simply the group of matrices themselves, acting on their natural vector space.

Example: $SU(2)$ fundamental representation

- **Representation Space (V):** A 2-dimensional complex vector space $\mathbb{C}^2$, physically corresponding to a spin-1/2 state

$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle. \quad (3.61)$$

- **Dimension of the representation:** $d = 2$ (since the matrices are 2×2). Note that the dimension of the $SU(2)$ group/algebra itself is $2^2 - 1 = 3$ (the number of

independent generators). The 3 Hermitian generators for this $d = 2$ fundamental representation are the Pauli matrices:

$$t^a = \frac{1}{2}\sigma^a \quad (a = 1, 2, 3), \quad (3.62)$$

which satisfy the Lie algebra $[t^a, t^b] = i\epsilon^{abc}t^c$, and the Pauli matrix identity:

$$\sigma^k \sigma^l = \delta^{kl} \mathbb{1} + i\epsilon^{klm} \sigma^m \quad (3.63)$$

Physically, we can use these fundamental building blocks to construct higher-dimensional representations via tensor products, such as the 3-dimensional spin-1 triplet state. In fact, $SU(2)$ has spin- j representations with dimension $2j + 1$. For instance, $j = 1/2$ corresponds to the 2-dimensional fundamental representation using 2×2 Pauli matrices.

- **Adjoint representation**

Instead of acting on physical states, the **adjoint representation** is a d -dimensional representation where the group/algebra acts on the Lie algebra space itself (spanned by the generators $\{t^a\}$ as basis vectors). The definition is as follows

Adjoint representation

We first define a automorphism map of group G :

$$\Psi : G \rightarrow \text{Aut}(G) \quad (3.64)$$

$$g \mapsto hgh^{-1} \equiv \Psi_h(g), \quad \forall h \in G, \quad (3.65)$$

where $\text{Aut}(G)$ is the automorphism group of G . Then we can define the derivative of Φ_g at the origin noted as Ad_g :

$$\text{Ad}_g = (d\Phi_g)_e : T_e G \rightarrow T_e G, \quad (3.66)$$

where $T_e G$ is the tangent space in the natural element of G . Thus the **adjoint representation** is a homomorphism:

$$\text{Ad} : G \rightarrow \text{Aut}(T_e G) \quad (3.67)$$

$$g \mapsto \text{Ad}_g. \quad (3.68)$$

To obtain the explicit matrix form of the adjoint representation, we consider an infinitesimal group element $g = e + i\alpha^a t^a \in G$. Acting on a generator $t^b \in T_e G$ via the adjoint map Ad_g , we expand to the linear order of α :

$$\text{Ad}_{e+i\alpha^a t^a}(t^b) = (e + i\alpha^a t^a)t^b(e - i\alpha^a t^a) \simeq t^b + i\alpha^a [t^a, t^b] + \mathcal{O}(\alpha^2) \quad (3.69)$$

This motivates us to define the infinitesimal adjoint map $\text{ad}_{t^a} \equiv d(\text{Ad})_e(t^a)$ on the Lie algebra, which acts purely as a commutator:

$$\text{ad}_{t^a}(t^b) = \lim_{\alpha^a \rightarrow 0} \frac{(\text{Ad}_{e+i\alpha^a t^a}(t^b))_{kl} - (t^b)_{kl}}{i\alpha^a} = [t^a, t^b]_{kl} = i \sum_{c=1}^d f^{abc} (t^c)_{kl}. \quad (3.70)$$

Viewing ad_{t^a} as a linear transformation matrix acting on the basis vector t^b to produce t^c , the expansion coefficients (structure constants) directly define the matrix elements of the adjoint generators:

$$(t_{\text{adj}}^a)_{bc} = i f^{abc} \quad (3.71)$$

Hence, the structure constants f^{abc} are not just algebraic constants, they are precisely **used to construct the adjoint representation**.

In this representation, the group elements act on the Lie algebra itself. The generators are $d \times d$ matrices (where $d = N^2 - 1$), with matrix elements defined by (3.71). We can verify that these matrices satisfy the Lie algebra by using the Jacobi identity:

$$\sum_{A=1}^d (f^{aAe} f^{bcA} + f^{bAe} f^{caA} + f^{cAe} f^{abA}) = 0. \quad (3.72)$$

Writing the commutation relation

$$[t_{\text{adj}}^a, t_{\text{adj}}^b]_{de} = i \sum_{c=1}^d f^{abc} (t_{\text{adj}}^c)_{de}, \quad (3.73)$$

in terms of structure constants gives:

$$\sum_{A=1}^d [(i f^{aAd})(i f^{bAe}) - (i f^{bAd})(i f^{aAe})] = i \sum_{c=1}^d f^{abc} (i f^{cde}). \quad (3.74)$$

Thus we have:

$$\sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe}) = - \sum_{A=1}^d f^{abA} f^{Ade}. \quad (3.75)$$

Using the anti-symmetry property of structure constant: $f^{abc} = -f^{bac}$, we can reduce to the standard Jacobi identity form:

$$0 = \sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} + f^{Ade} f^{abA}) \quad (3.76)$$

$$= \sum_{A=1}^d (f^{aAd} f^{beA} + f^{bAd} f^{eaA} + f^{eAd} f^{abA}), \quad (3.77)$$

which is nothing but the Jacobi identity (3.72).

1.5 Inner product of generators

Now since the Lie algebra generators are all matrices, so it's natural to consider the inner product of two matrices. The most natural way is by the trace of matrix product. So we define a matrix D_r^{ab} for generators t_r^a as

$$D_r^{ab} = \text{tr}(t_r^a t_r^b). \quad (3.78)$$

For $a, b = 1, \dots, d = N^2 - 1$. Cyclicity of trace doesn't change the matrix, so it's symmetry and real-valued:

$$D_r^{ab} = D_r^{ba}, \quad (3.79)$$

and also Hermitian:

$$(D_r^{ab})^\dagger = \text{tr}((t_r^a)^\dagger (t_r^b)^\dagger) = D_r^{ab}. \quad (3.80)$$

As we know from linear algebra, for any real-valued symmetry matrix, one can always diagonalize it by a orthogonal transformation. Thus, diagonalization of D_r^{ab} gives

$$\text{tr}(t_r^a t_r^b) = C(r) \delta^{ab}. \quad (3.81)$$

Since the generators are diagonal, then the position of each components in the same t_r are equal, which imply that $C(r)$ is the same for any component of t_r . Thus $C(r)$ is called the **index of representation** or **Dynkin index**.

Example: $SU(N)$

The generators for $SU(2)$ is $t_{\text{fund}}^a = \frac{\sigma^a}{2}$ ($a = 1, 2, 3$), where σ^a is the Pauli matrix:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.82)$$

For the fundamental representation matrix, we can check that

$$\text{tr}(t_r^a t_r^a) = \frac{1}{2}. \quad (3.83)$$

So

$$C(r) = \frac{1}{2}. \quad (3.84)$$

More over, for general case $SU(N)$, $C(r) = \frac{1}{2}$ due to the normalization of N -dim matrices are all similar with $SU(2)$. Using the index of representation, we can find the structure constants of $SU(N)$ satisfy:

$$\begin{aligned} \text{tr}([t^a, t^b] t_r^c) &= i f^{abc} \text{tr}(t_r^c t_r^c) \\ &= \text{tr}(t_r^a t_r^b t_r^c - t_r^b t_r^a t_r^c) = i f^{abc} C(r) \\ &= \text{tr}(t_r^a [t_r^b, t_r^c]) = i f^{abc} C(r). \end{aligned} \quad (3.85)$$

This shows that f^{abc} are totally anti-symmetric:

$$f^{abc} = -f^{bac} = f^{bca}. \quad (3.86)$$

As we all learned from quantum mechanics: $[J^2, J_i] = 0$. So similarly, one can define an operator T^2 that satisfies the similar commutation relation. This operator is called the **quadratic Casimir operator**, which is defined as:

$$T_{ij}^2 = \sum_{a=1}^{d(G)} \sum_{k=1}^{d(r)} t_{ik}^a t_{kj}^a, \quad (3.87)$$

where t_{ij}^a is fundamental representation. We can check that for any a , we all have: $[t^a, T^2] = 0$, namely T^2 commutes with all generators:

$$[t^a, T^2] = \left[t^a, \sum_b t^b t^b \right] = \sum_b ([t^a, t^b] t^b + t^b [t^a, t^b]) \quad (3.88)$$

$$= \sum_b \left(\left(i \sum_c f^{abc} t^c \right) t^b + t^b \left(i \sum_c f^{abc} t^c \right) \right) \quad (3.89)$$

$$= i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.90)$$

$$= i \sum_{c,b} f^{acb} (t^b t^c + t^c t^b) \quad (3.91)$$

$$= i \sum_{b,c} (-f^{abc}) (t^c t^b + t^b t^c) \quad (3.92)$$

$$= -i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.93)$$

$$= 0 \quad (3.94)$$

This implies that T^2 must be proportional to identity:

$$T_{ij}^2 = C_2(r) \delta_{ij} \quad (3.95)$$

And $C_2(r)$ is called the **quadratic Casimir coefficient**. It can be obtained by taking the trace in both sides:

$$\text{tr}(T^2) = \text{tr} \left(\sum_{a=1}^{d(G)} t^a t^a \right) = \sum_{a=1}^{d(G)} \text{tr}(t^a t^a) = C_2(r) \text{tr} \delta_{ij}, \quad (3.96)$$

thus we have:

$$C(r) d(G) = C_2(r) d(r) \quad (3.97)$$

1.6 Representation of Lorentz group

The Lorentz group, as we mentioned before, is $SO(1, 3)$ with metric $\eta = (1, -1, -1, -1)$. The group element Λ with infinitesimal expansion can be written as

$$\Lambda_\nu^\mu = \delta_\nu^\mu + i \sum_a \omega^a (t^a)_\nu^\mu + \mathcal{O}(\omega^2). \quad (3.98)$$

Defining relation of $SO(N, M)$ groups says

$$\Lambda^T \eta \Lambda = \eta, \quad (3.99)$$

thus the infinitesimal form implies

$$(\mathbb{1} + i\omega^a (t^a)^T) \cdot \eta \cdot (\mathbb{1} + i\omega^a t^a) = \eta, \quad (3.100)$$

after dropping the $\mathcal{O}(\omega^2)$, one obtains the anti-symmetric relation:

$$(t^a)^T \eta = -\eta t^a. \quad (3.101)$$

As we mentioned in the last subsection, $SO(1, 3)$, as a subgroup of $O(1, 3)$, has $\frac{1}{2} \times 4 \times (4-1) = 6$ generators, physically are 3 rotation + 3 boosts, which usually noted as $J^{\mu\nu}$. To limit the number of free parameters in $J^{\mu\nu}$, we require

$$J^{\rho\sigma} = -J^{\sigma\rho}. \quad (3.102)$$

So we change the note as

$$\sum_a \omega^a t^a \rightarrow \sum_{\rho\sigma} \omega_{\rho\sigma} J^{\rho\sigma}, \quad (3.103)$$

where $J^{\rho\sigma}$ fulfill $SO(1, 3)$ algebra. Generally, all $SO(N, M)$ groups have also spinor representations generators

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (3.104)$$

fulfill $SO(1, 3)$ algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}). \quad (3.105)$$

1.7 Integration over gauge group

Since a Lie group is also a manifold, so we can introduce the integration structure based on the group element g and function $f(g)$, the measurement dg is called Haar measure, which include the left and right invariant measure:

$$\text{left invariant measure : } \int (dg)_L f(g) = \int (dg)_L f(g'g); \quad (3.106)$$

$$\text{right invariant measure : } \int (dg)_R f(g) = \int (dg)_R f(gg'), \quad (3.107)$$

where the normalization is $\int dg \, 1 = 1$, and for most of situation the left and right Haar measure are the same $dg = (dg)_L = (dg)_R$. This measurement is just like the normal integral is invariant under the transformation $x \rightarrow x + C$, where C is a constant:

$$\int dx f(x) = \int dx f(x + C). \quad (3.108)$$

To obtain the computable form of the measure, we need explicit realization of dg with metric tensor M_{ij} :

$$M_{ij} \equiv \text{tr}(g^{-1} \partial_i g \cdot g^{-1} \partial_j g), \quad (3.109)$$

where $\partial_i g = \frac{\partial}{\partial \alpha_i} g(\alpha)$. Then we can connect to the normal integral:

$$\int dg f(g) = N \int d\alpha |\det M|^{1/2} f(g(\alpha)), \quad (3.110)$$

in which the normalization is the Jacobi determinant.

Example

- $U(1) = \{e^{i\theta} \mid -\pi < \theta < \pi\}$: For $g(\theta) = e^{i\theta}$ we have

$$M_{\theta\theta} = \text{tr}(g^{-1} \partial_\theta g \cdot g^{-1} \partial_\theta g) = \text{tr}(i \cdot i) = -1. \quad (3.111)$$

And using the fact

$$1 = N \int_{-\pi}^{\pi} d\theta \cdot 1 = N \cdot 2\pi, \quad (3.112)$$

thus $N = \frac{1}{2\pi}$. So the integral is

$$\int dg f(g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta f(e^{i\theta}) \quad (3.113)$$

- $SU(2) = \{a_0 + i\vec{\sigma} \cdot \vec{a} \mid a_0^2 + \vec{a}^2 = 1\}$: For the element $g(a)$ we have:

$$\partial_0 g = \frac{\partial g}{\partial a_0} = \mathbb{1}_{2 \times 2} \quad (3.114)$$

$$\partial_k g = \frac{\partial g}{\partial a_k} = i\sigma_k \quad (k = 1, 2, 3) \quad (3.115)$$

Thus, the metric is

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}, \quad (3.116)$$

therefore, $|\det M|^{1/2} = \sqrt{16} = 4$. The last piece is the normalization factor, for that we need to compute the integral for unity. Since the constrain require a sphere, we then can express the unity by delta function:

$$1 = N \int d^4a \delta(a^2 - 1) = N \int_0^\infty r^3 dr \int d\Omega_3 \delta(r^2 - 1), \quad (3.117)$$

thus $N = \frac{1}{\pi^2}$. And finally:

$$\int dg f(g) = \frac{1}{\pi^2} \int d^4a \delta(a^2 - 1) f(g(a)). \quad (3.118)$$

- More general case, for $SU(N)$, Schur's orthogonality relation requires

$$\int dg g_{ij}(g^{-1})_{kl} = \frac{1}{N} \delta_{il} \delta_{jk}. \quad (3.119)$$